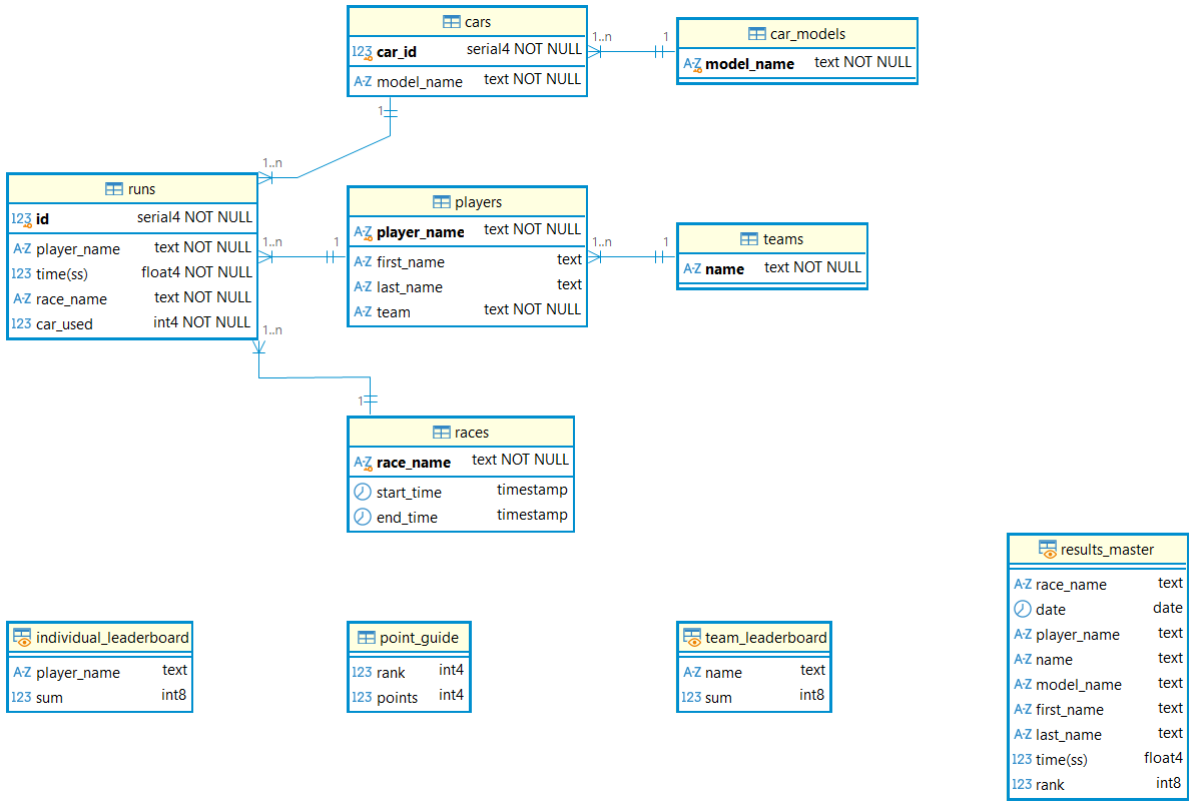
ER Diagram



While conceptually the cars – models, runs-players, runs-cars relationships would be [0:n]-[1:1], DBeaver displays them as [1:n]-[1:1]. Runs-races have [2:n]-[1:1] cardinality – this rule I could not enforce into the modelling part, however it is utilized when team results are filtered in the team_leaderboard.

team_leaderboard

	name	sum
1	Falcons	14
2	Wolves	11
3	Panthers	9
4	Tigers	4

```sql
CREATE VIEW team_leaderboard AS
SELECT teams.name, SUM(points)
FROM teams
    LEFT JOIN results_master USING (name)
    LEFT JOIN point_guide USING (rank)
GROUP BY teams.name
HAVING teams.name IN(
    SELECT team
    FROM players
);
```

individual_leaderboard

	player_name	sum
1	bence_nagy	8
2	lena_muller	8
3	marta_silva	4
4	jonas_weber	3
5	david_kim	5
6	anna_kovacs	6
7	sofia_rossi	4
8	noah_stein	[NULL]

```sql
CREATE VIEW individual_leaderboard AS
SELECT players.player_name, sum(points)
FROM players
    LEFT JOIN results_master USING
(player_name)
    LEFT JOIN point_guide USING (rank)
GROUP BY players.player_name;
```

results_master view

```sql
CREATE VIEW results_master AS
SELECT
    race_name,
    DATE(races.start_time),
    runs.player_name,
    name,
    model_name,
    first_name,
    last_name,
    runs."time(ss)",
    RANK() OVER (
        PARTITION BY race_name
        ORDER BY runs."time(ss)" ASC
    ) AS rank
FROM races
INNER JOIN runs USING (race_name)
LEFT JOIN cars ON runs.car_used = cars.car_id
LEFT JOIN car_models USING (model_name)
LEFT JOIN players USING (player_name)
LEFT JOIN teams ON teams.name = players.team
WHERE race_name IN (
    SELECT race_name
    FROM runs
    GROUP BY race_name
    HAVING COUNT(DISTINCT player_name) >= 2
);
-- a race is displayed only if there are at least two players;
```